

# External Validation One-Page

MIDAS Section Library

embedded metadata validated

static + contract gate

MIDAS section-library: ok | 182/183 used | 183 templates | source=midas\_parser\_derived |  
implementation/phase1/open\_data/midas/midas\_generator\_33.json  
nightly, release gap, committee dashboard, and this external validation onepage all consume the same validator line

Validator line

MIDAS section-library: ok | 182/183 used | 183 templates | source=midas\_parser\_derived |  
implementation/phase1/open\_data/midas/midas\_generator\_33.json

|                      |                                                         |
|----------------------|---------------------------------------------------------|
| Bundle               | 20260330T150627Z                                        |
| Generated            | 2026-03-30T15:06:27.956197+00:00                        |
| Nightly              | PASS                                                    |
| CI                   | PASS                                                    |
| Static validation    | PASS                                                    |
| Freeze               | PASS                                                    |
| Promotion            | PASS                                                    |
| Promotion reason     | PASS                                                    |
| Promotion hold       | False                                                   |
| Hold review manifest | implementation/phase1/release/hold_review_manifest.json |
| Hold review packet   | implementation/phase1/release/hold_review_packet.md     |

# External Validation One-Page (Continued)

|                                 |                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------|
| Hold review packet pdf          | implementation/phase1/release/hold_review_packet.pdf                                           |
| Hold review ack                 | implementation/phase1/release/hold_review_ack.json                                             |
| Smoke reason                    | PASS                                                                                           |
| Smoke pass rate                 | 100.00%                                                                                        |
| Smoke trial feasible            | 100.00%                                                                                        |
| Smoke avg trial runtime (s)     | 0.0557                                                                                         |
| Smoke strict recommendation     | candidate_for_strict_enable                                                                    |
| Signed registry                 | PASS                                                                                           |
| Registry signature              | PASS                                                                                           |
| Solver HIP e2e                  | PASS                                                                                           |
| RC benchmark lock               | PASS                                                                                           |
| NDTHA residual gate             | PASS                                                                                           |
| Committee package               | PASS                                                                                           |
| MIDAS section-library validator | MIDAS section-library: ok   182/183 used   183 templates   source=midas_parser_derived   imple |
| MIDAS section-library status    | embedded metadata validated                                                                    |
| Material constitutive           | Material constitutive gate: PASS   concrete_damage=yes(matrix=48/48,max=1.000)   cyclic_deg    |
| Surface interaction benchmark   | Surface interaction benchmark: PASS   ready=7/7   family_matrix=420/420   source_families=10   |
| KDS row provenance export       | MIDAS KDS row provenance export: PASS   combos=6   rows=144   members=12   clauses=6   e       |

# External Validation One-Page (Continued)

|                                          |                                                                                                |
|------------------------------------------|------------------------------------------------------------------------------------------------|
| Commercial grade                         | Commercial                                                                                     |
| Deployment model                         | engineer_in_the_loop_accelerated_coverage                                                      |
| Accelerated coverage                     | 95-99%                                                                                         |
| Residual holdout                         | 1-5%                                                                                           |
| Estimated time saved                     | 91-96%                                                                                         |
| Empirical smoke runtime reduction        | 95.9-96.23%                                                                                    |
| Comparable run mode                      | current_pipeline_comparable_full_chain_pass                                                    |
| Comparable deployment                    | engineer_in_the_loop_accelerated_coverage                                                      |
| Comparable strict smoke                  | True                                                                                           |
| Engineer-in-loop ready                   | True                                                                                           |
| Estimated time basis                     | Empirical estimate derived from nightly design-optimization smoke runtime reduction, scaled by |
| Time-saving focus                        | Use this engine to automate the dominant, time-consuming 95-99% of repeated analysis, screen   |
| Full replacement ready                   | False                                                                                          |
| External benchmark start now             | True                                                                                           |
| External benchmark full submission ready | False                                                                                          |
| External benchmark reason                | PASS_START_NOW_LIMITED                                                                         |
| External benchmark start mode            | start_now_limited_external_benchmark                                                           |
| External benchmark scope                 | component_and_system_performance_benchmark_with_review_boundary                                |

# External Validation One-Page (Continued)

|                                               |                                                                                               |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------|
| External benchmark blockers                   | none                                                                                          |
| External benchmark cautions                   | panel_zone_external_validation_only_boundary, audit_review_queue_pending=2                    |
| External benchmark execution mode             | limited                                                                                       |
| External benchmark execution ready tasks      | 10                                                                                            |
| External benchmark execution blocked tasks    | 2                                                                                             |
| External benchmark execution review-boundary  | 2; pending                                                                                    |
| External benchmark review-boundary resolution | approve_all=PASS_START_NOW_FULL/ready_full=yes; reject_one=ERR_ARCHITECTURE_BLOCKERS          |
| External benchmark execution status           | execution_complete_no_fail                                                                    |
| External benchmark execution status counts    | planned=0   in_progress=0   completed=10   failed=0   finished=10   completion_ratio=1.000    |
| PBD response source                           | resolved_report=implementation/phase1/experiments/by_test/nonlinear_ndtha_stress/2026031      |
| Audit review decision batch template          | items=2   status=pending_review=2   owner=licensed_engineer=2   priority=high=1, medium=1     |
| Approve-all readiness preview                 | reason=PASS_START_NOW_FULL   ready_full=True   pending=0   open_revision=0                    |
| Reject-one readiness preview                  | reason=ERR_ARCHITECTURE_BLOCKERS   ready_full=False   pending=0   open_revision=1   blocked   |
| Audit review decision batch runner            | reason=PASS   apply_live=False   live_applied=False   preview_reason=PASS_START_NOW_FULL      |
| Structural optimization viewer                | implementation/phase1/release/visualization/structural_optimization_viewer.html   mode=static |
| Authority routing warning                     | none                                                                                          |
| Open gaps                                     | P0=1 / P1=0 / P2=0                                                                            |
| MIDAS rows                                    | total=12728, skipped=0, unknown=0                                                             |

# External Validation One-Page (Continued)

|                                            |                                                                                             |
|--------------------------------------------|---------------------------------------------------------------------------------------------|
| KDS rows                                   | compliance=511, member_checks=1056, clauses=16                                              |
| NDTHA residual                             | top=0.6880580090188355 m, drift=1.9136000000000006 %, fallback_rate=0.0                     |
| Registry artifacts                         | 9                                                                                           |
| Design-opt long feasible                   | True                                                                                        |
| Design-opt long final max DCR              | 0.9331636363636363                                                                          |
| Design-opt raw max drift                   | 9.200000000000003                                                                           |
| Design-opt repaired compliance max drift   | 1.9872000000000005                                                                          |
| Design-opt compliance basis                | repaired_solver_validated_slice                                                             |
| Design-opt repair action count             | 1                                                                                           |
| Design-opt constructability signal gain    | 0.8158195424350374                                                                          |
| Design-opt constructability avg            | 0.32955670560576916 -> 0.3268681175980322                                                   |
| Design-opt detailing complexity avg        | 0.44354114999544225 -> 0.4417926006724248                                                   |
| Design-opt selected family mix             | beam_section=4, connection_detailing=6, detailing=5, perimeter_frame=1, rebar=5, slab_thick |
| Design-opt selected dominant family        | wall_thickness (21.88%)                                                                     |
| Design-opt selected family mix trend       | beam_section=+0, connection_detailing=+0, detailing=+0, perimeter_frame=+0, rebar=+0, slab  |
| Design-opt previous dominant family        | wall_thickness (21.88%)                                                                     |
| Design-opt preview supply family mix       | beam_section=8, connection_detailing=9, detailing=164, perimeter_frame=1, rebar=276, slab   |
| Design-opt preview missing target families |                                                                                             |

# External Validation One-Page (Continued)

|                                         |                                                                                             |
|-----------------------------------------|---------------------------------------------------------------------------------------------|
| MGT export direct-patch changes         | 25                                                                                          |
| MGT export direct-patch families        | beam_section=1, connection_detailing=6, detailing=5, perimeter_frame=1, rebar=5, slab_thick |
| MGT rebar namespace mode                | group_local                                                                                 |
| MGT rebar delivery mode                 | direct_patch_eligible                                                                       |
| MGT evidence model                      | direct_patch_plus_audit_review_manifest                                                     |
| MGT rebar material namespace present    | True                                                                                        |
| MGT rebar group-local namespace present | True                                                                                        |
| MGT material rebar payloads             | 3/5                                                                                         |
| MGT group-local rebar payloads          | 6                                                                                           |
| MGT rebar direct-patch eligible         | 6                                                                                           |
| MGT rebar direct-patch blockers         |                                                                                             |
| MGT rebar mapping sources               | alt_slab_wall_group_id=5, direct_group_id=1                                                 |
| Design-opt cost delta                   | 903.4629446021863                                                                           |
| Design-opt changed groups               | 25                                                                                          |
| Blocked cost-down rows                  | 50                                                                                          |
| Blocked illegal-by-mask                 | 0                                                                                           |
| Illegal-by-mask families                |                                                                                             |
| Blocked no-cost-gain                    | 9                                                                                           |

# External Validation One-Page (Continued)

|                                    |                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------|
| Blocked constructability hard-gate | 5                                                                                      |
| Hard-gate reasons                  | detailing_ratio_above_hard_limit=5                                                     |
| Hard-gate families                 | beam_section=2, rebar=3                                                                |
| No-cost-gain groups                | 5                                                                                      |
| No-cost-gain explain rows          | 5                                                                                      |
| Design-opt groups                  | Ablation=1/1   Dataset=1/1   Profile=1/1   Stage A=2/2   Stage A/B/C=1/1   Stage B=1/1 |
| Frame metrics                      | cases=3, drift_p95=4.167%, top_disp_p95=3.485%                                         |
| Wind metrics                       | cases=4, max_drift=0.000806%, residual_drift=0.000113%                                 |
| SSI metrics                        | cases=4, nonlinear_span=0.282342, residual_drift=0.004034%                             |
| Design-opt metrics                 | rows=17, raw_drift=9.2000%, repaired_drift=1.9872%, cost_delta=903.463                 |

This is the current external-validation submission baseline. Previous submission bundles were pruned after this package was created.

# Appendix: MIDAS KDS Row Provenance Export

MIDAS KDS row provenance export: PASS | combos=6 | rows=144 | members=12 | clauses=6 | exact\_rows=144 | clause\_filters=6 | member\_filters=12 | reverse\_jump=viewer\_subset\_reverse\_jump\_v11

json=implementation/phase1/release/kds\_compliance/midas\_kds\_row\_provenance\_table.json |  
csv=implementation/phase1/release/kds\_compliance/midas\_kds\_row\_provenance\_table.csv |  
report=implementation/phase1/release/kds\_compliance/midas\_kds\_row\_provenance\_table\_report.json

gLCB1 | member=C-TST-003 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

gLCB1 | member=C-TRN-005 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member\_types=column

gLCB1 | member=C-TST-003 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

gLCB1 | member=C-TST-001 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-001 | case=C-TST-001 | baseline=27441 | member\_types=column

gLCB1 | member=C-TST-003 | clause=KDS-SHEAR-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

gLCB1 | member=C-TRN-005 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member\_types=column

# Appendix: MIDAS KDS Row Provenance Export

gLCB1 | member=C-TRN-007 | clause=KDS-MOMENT-Y-001 | baseline=27425

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-007 | case=C-TRN-007 | baseline=27425 | member\_types=wall

gLCB1 | member=C-TST-003 | clause=KDS-AXIAL-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

clause=KDS-AXIAL-001 | rows=24 | members=12 | combos=2 | top\_member=C-TST-003 | top\_dcr=1.065

clause=KDS-INT-FRAME-001 | rows=24 | members=12 | combos=2 | top\_member=C-TST-003 | top\_dcr=1.137

clause=KDS-MOMENT-Y-001 | rows=24 | members=12 | combos=2 | top\_member=C-TST-003 | top\_dcr=1.216

clause=KDS-MOMENT-Z-001 | rows=24 | members=12 | combos=2 | top\_member=C-TST-003 | top\_dcr=0.991

clause=KDS-SHEAR-Y-001 | rows=24 | members=12 | combos=2 | top\_member=C-TST-003 | top\_dcr=1.110

clause=KDS-SHEAR-Z-001 | rows=24 | members=12 | combos=2 | top\_member=C-TST-003 | top\_dcr=0.863

member=C-TRN-001 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=C-TRN-002 | baseline=27287 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=C-TRN-003 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=C-TRN-004 | baseline=27425 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=C-TRN-005 | baseline=27441 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=C-TRN-006 | baseline=27287 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

## Appendix: MIDAS KDS Row Provenance Export

member=C-TRN-007 | baseline=27425 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=C-TST-001 | baseline=27441 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

hazard=combined | rows=48 | members=4 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=1.216

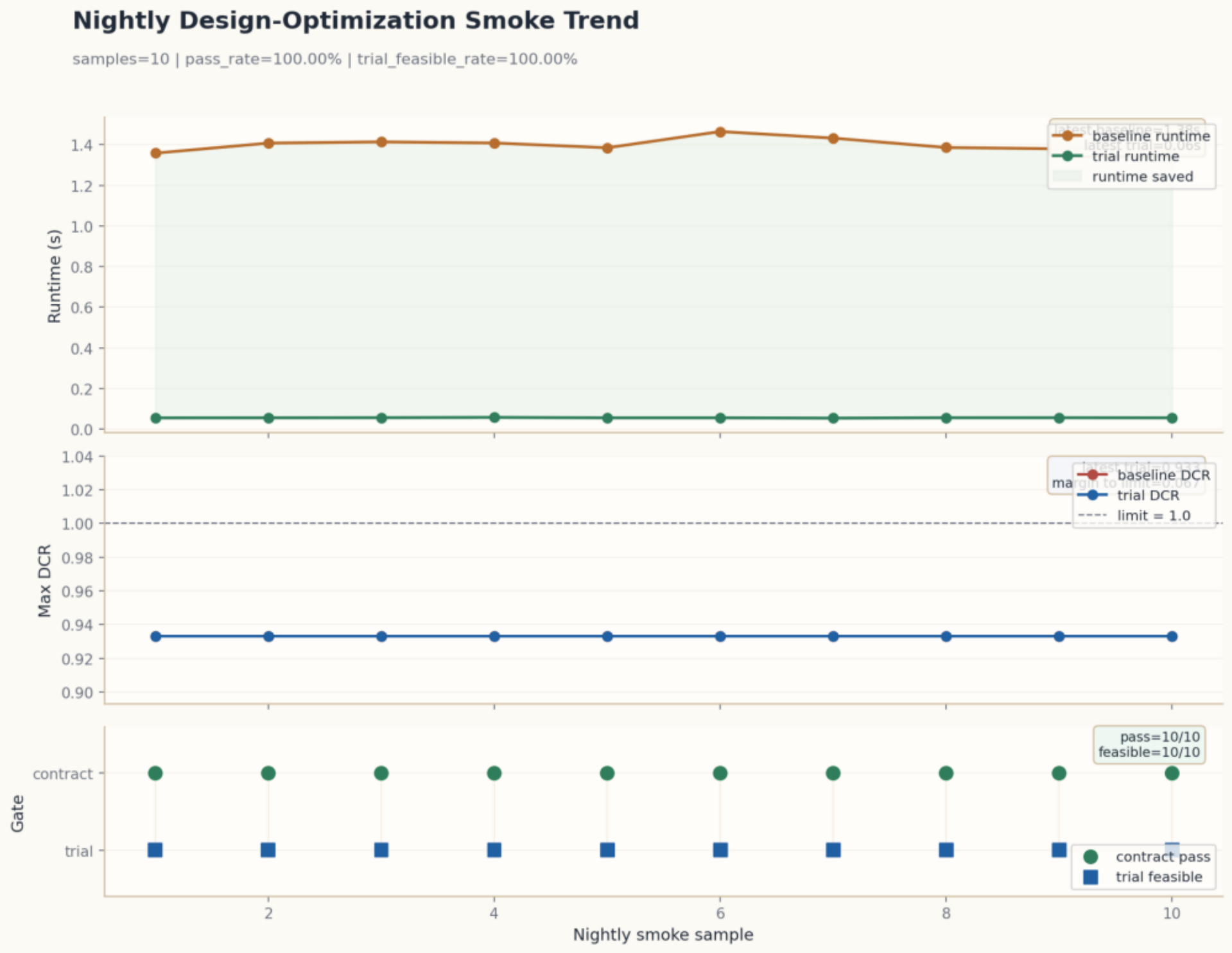
hazard=seismic | rows=48 | members=4 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=1.112

hazard=wind | rows=48 | members=4 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=0.809

rule\_family=strength | rows=120 | members=12 | hazards=3 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=1.216

rule\_family=strength\_interaction | rows=24 | members=12 | hazards=3 | combos=2 | top\_clause=KDS-INT-FRAME-001 | top\_dcr=1.137

# Nightly Smoke Trend



# Nightly Smoke Recent Samples

#6 2026-03-30T13:33:18.377052+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.4648s | trial\_runtime=0.0555s | trial\_max\_dcr=0.9332 | action=connect

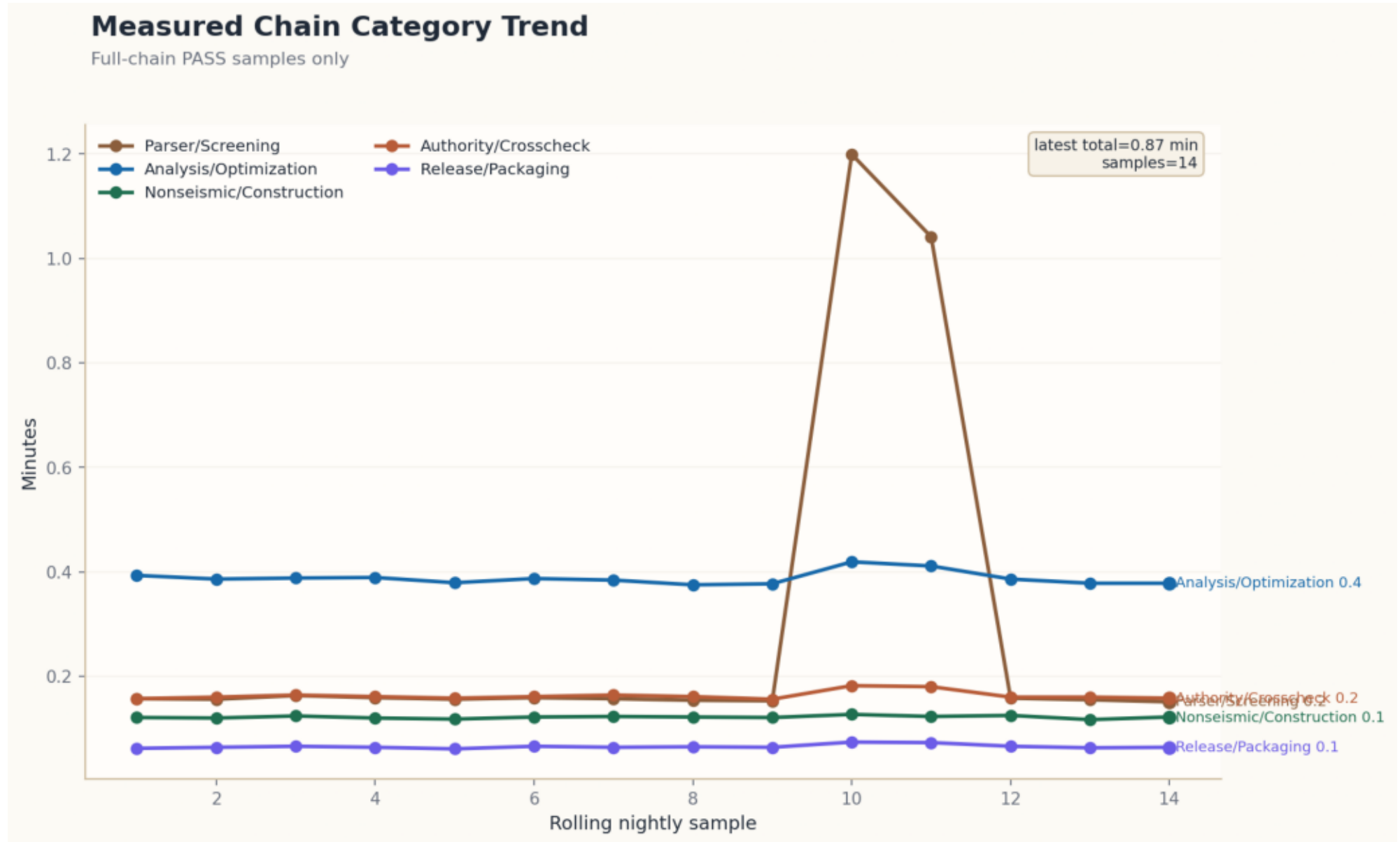
#7 2026-03-30T13:57:14.403720+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.4324s | trial\_runtime=0.0540s | trial\_max\_dcr=0.9332 | action=connect

#8 2026-03-30T13:58:15.190727+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.3860s | trial\_runtime=0.0559s | trial\_max\_dcr=0.9332 | action=connect

#9 2026-03-30T14:33:48.013233+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.3798s | trial\_runtime=0.0560s | trial\_max\_dcr=0.9332 | action=connect

#10 2026-03-30T14:37:33.867618+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.3774s | trial\_runtime=0.0555s | trial\_max\_dcr=0.9332 | action=connect

# Measured Chain Category Trend



# Residual Holdout Boundary

Licensed Engineer Review | owner=기술사 | share=50% | project=0.5-2.5% | non-standard interpretation, final judgment, exceptional irregularity, and member-level ed

Legacy Tool Cross-Validation | owner=기존툴+기술사 | share=30% | project=0.3-1.5% | novel load paths, authority-critical submodels, and residual niche workflows out

Legal Sign-Off | owner=기술사/기존 승인 workflow | share=20% | project=0.2-1.0% | formal seal, legal submission, and authority-facing responsibility that remains out

# Time-Saving Coverage

Coverage target: 95-99%

Residual holdout: 1-5%

Estimated time saved: 91-96%

Measured chain wall-clock (comparable rolling): 1.03 min

Rolling sample count: 14, range=0.87-2.0 min

Measured chain wall-clock (current): 0.87 min

Comparable run mode: `current_pipeline_comparable_full_chain_pass`

Empirical smoke runtime reduction: 95.9-96.23%

Basis: Empirical estimate derived from nightly design-optimization smoke runtime reduction, scaled by the accelerated-coverage target.  
`smoke_mean_runtime_saved=96.02%`, `sample_count=10`.

Use this engine to automate the dominant, time-consuming 95-99% of repeated analysis, screening, packaging, and optimization workflows. Keep the residual 1-5% under licensed engineer review, legacy-tool cross-check, and formal sign-off workflows.

# Residual Holdout Review Table

Licensed Engineer Review | review\_story\_zone | S04/perimeter (2), S01/intermediate (1), S02/perimeter (1), S03/intermediate (1)

owner=기술사 | why=Top story-zone review pockets are derived from actual accepted design-change rows so engineer holdout stays tied to the highest-touch parts of the structure.

Licensed Engineer Review | story\_band | 4, 3, 6

owner=기술사 | why=High-touch story bands remain under engineer review because they concentrate accepted design changes and irregular response checks.

Licensed Engineer Review | member\_family | wall (28), slab (20), beam (20)

owner=기술사 | why=Dominant member families in accepted optimization changes still require engineer judgment on local edge cases and detailing intent.

Licensed Engineer Review | zone | perimeter (9), intermediate (6), core (1)

owner=기술사 | why=Zone concentration is used to focus manual review on the highest-touch portions of the structural layout.

Legacy Tool Cross-Validation | submodel\_family | SCBF16B (3), SCBF16B\_shell\_beam\_mix (2), nheri\_case01\_sensor (1), nheri\_case02\_sensor (1)

owner=기존툴+기술사 | why=Authority submodel families are derived from the active catalog paths so cross-validation follows the exact benchmark submodels still outside the accelerated envelope.

Legacy Tool Cross-Validation | authority\_critical\_case | SAC (3), NHERI (3), OpenSees (2)

owner=기존툴+기술사 | why=Authority-critical benchmark tracks remain the primary cross-validation target outside the accelerated envelope.

Legacy Tool Cross-Validation | authority\_catalog\_case\_id | SAC20\_LA\_holdout\_01, SAC20\_SEA\_holdout\_02, SAC20\_BOS\_holdout\_03, NHERI\_UCSD\_like\_holdout

owner=기존툴+기술사 | why=Authority catalog case ids are read directly so the holdout review list refreshes automatically when the benchmark catalog changes.

Legal Sign-Off | authority\_catalog\_track | sac (3), nheri (3), opensees (2)

owner=기술사/기존 승인 workflow | why=Formal authority-facing responsibility is anchored to the active authority catalog tracks and remains outside the automated responsibility boundary.

Legal Sign-Off | authority\_critical\_case | sealed submission pack, authority-facing variants, stamped final issue

owner=기술사/기존 승인 workflow | why=Formal authority-facing deliverables remain outside the automated responsibility boundary.

# Residual Holdout Routing Matrix

Legacy Tool Cross-Validation | track=opensees | submodel=SCBF16B

review=S04/perimeter | member=wall | owner=기존툴+기술사

why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

Legacy Tool Cross-Validation | track=opensees | submodel=SCBF16B\_shell\_beam\_mix

review=S01/intermediate | member=slab | owner=기존툴+기술사

why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

Legacy Tool Cross-Validation | track=sac | submodel=SCBF16B

review=S02/perimeter | member=beam | owner=기존툴+기술사

why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

Legacy Tool Cross-Validation | track=sac | submodel=SCBF16B\_shell\_beam\_mix

review=S03/intermediate | member=column | owner=기존툴+기술사

why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

Legacy Tool Cross-Validation | track=nheri | submodel=nheri\_case01\_sensor

review=S04/perimeter | member=wall | owner=기존툴+기술사

why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

Legacy Tool Cross-Validation | track=nheri | submodel=nheri\_case02\_sensor

review=S01/intermediate | member=slab | owner=기존툴+기술사

why=Routing matrix links active authority submodels to dominant story/zone/member-family review pockets so the remaining manual and legacy-tool work stays explicit.

# Authority Catalog Routing Diff

baseline\_seeded=False | changes=0 | added=0 | removed=0 | unchanged=7

No authority-catalog routing changes detected for this external submission refresh.